

W07 MOISTURE MANAGEMENT | O (MAX : 3 PT)

Intent : Limit the potential for bacteria and mold growth within buildings from water infiltration, condensation and internal leaks.

Summary : This WELL feature requires projects to develop strategies to minimize the presence of unintentional water and, when unavoidable, to manage it through material selection and inspections.

Issue : Excess moisture and dampness is a common problem in buildings, affecting about 20% of buildings in Europe, Canada and the U.S.¹ When improperly managed, moisture creates conditions conducive to the growth of mold and other biological pests, which can increase the risk of developing respiratory infections and asthma for those within the building.² It is estimated that one fifth of asthma cases in the U.S. are caused by excess moisture and dampness in buildings.² Furthermore, moisture can damage the building itself by creating an environment hospitable to insects and other destructive pests, corroding metal components and degrading wood and porous building materials.³ These problems can arise when water unintentionally penetrates the building envelope or leaks from indoor uses of water, or when moisture-heavy air condenses on building materials.

Solutions : Through effective design of the building's curtain wall, water piping assemblies and ventilation systems, and by selecting appropriate materials where condensation may occur, projects can make conditions inhospitable to mold, microbes and pests, reducing the risk to respiratory health.^{3,4} Inspections are needed to both verify that design and operations properly safeguards against mold growth, as well as inform the need for preventative maintenance.⁴

PART 1 DESIGN ENVELOPE FOR MOISTURE PROTECTION (MAX : 1 PT)

For All Spaces:

The building envelope aims to minimize moisture intrusion and accumulation through the following:³

- a. For a project where construction occurs after enrollment or the start of subscription, verification of site drainage and storm water management during building construction phase.
- b. Air tightness testing to assess water vapor transfer.
- c. Adverse vapor pressure differentials that may cause condensation on interstitially hidden materials.
- d. Entryway design that considers at least three strategies to minimize the ingress or permeation of water into the building.
- e. Installation of a continuous drainage plane (e.g., a weather-resistant barrier integrated with flashing systems at penetrations), interior to the exterior cladding.
- f. Minimization of capillary suction (wicking) in porous building materials through one of the below capillary break methods:
 1. Free-draining spaces (e.g., between exterior cladding, weather-resistant barriers in wall assemblies).
 2. Non-porous materials (e.g., closed-cell foams, waterproofing membranes, metal) between porous materials.

PART 2 DESIGN INTERIORS FOR MOISTURE MANAGEMENT (MAX : 1 PT)

For All Spaces:

1: Condensation and liquid water management

The project implements measures to manage moisture in interior spaces that address, at a minimum, the following:³

- a. Protection of moisture-sensitive building materials and selection of moisture-resistant materials or finishes in surfaces likely to be exposed to liquid water (e.g., finished floors) or that may absorb moisture such as interior sheathing in basements, areas at or below grade, bathrooms, janitorial rooms or kitchens.
- b. Condensation on cold surfaces such as basements, slab-on-grade floors, the inside of exterior walls and glazing.

2: Water leak control in fixtures

The following requirements are met:

- a. All hard-piped fixtures, such as toilets, dishwashers, icemakers, water treatment devices and clothes washers, have a labeled, readily accessible single-throw manual shut-off or automatic shut-off at point-of-connection.
- b. For water treatment devices that have a waste or drain line (e.g., reverse osmosis systems and water softeners), the drain or waste line is plumbed in-place and is equipped with a backflow prevention system such as an air gap or a backflow preventer valve.

PART 3 IMPLEMENT MOLD AND MOISTURE MANAGEMENT PLAN (MAX : 1 PT)

For All Spaces:

1: Operational moisture management

The project implements a moisture management plan for building operations that contains the following:

- a. A schedule of periodic inspections for signs and potential sources of water damage or pooling, discoloration and mold on ceilings, walls, floors and HVAC equipment.³
- b. A system or inspection protocol to periodically assess water pipe leaks.
- c. A system for occupants and tenants to notify building management about mold or water damage.

2: Leaks and mold inspections

The following requirement is met:

- a. Results of inspections for mold and leaks (including any mold test results) are submitted annually through the WELL digital platform.

REFERENCES

1. Institute of Medicine. Damp Indoor Spaces and Health. Washington, D.C.: National Academies Press; 2004.
2. Mudarri D, Fisk WJ. Public health and economic impact of dampness and mold. Indoor Air. 2007;17(3).
3. U. S. Environmental Protection Agency. Moisture Control Guidance for Building Design, Construction and Maintenance. Washington, DC2013.
4. World Health Organization. WHO guidelines for indoor air quality: dampness and mould. 2009.